

Microneedle biomedical devices

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Abstract

Microneedle (MN) technologies have gained interest for biomedical applications, ranging from drug delivery to sensing and tissue sampling in skin and internal organs. The microscale feature of MN systems allows the accommodation of therapeutics from small drugs to macromolecule biologics and even living cells. Additionally, it permits body fluid extraction when inserted into tissues. The integration of MNs with other technologies, such as stimuli-responsive materials, smart electronics and wearables, facilitates precision drug delivery, dynamic monitoring and closed-loop theranostic functions. In this Review, we discuss MN materials, structures and engineering approaches, focusing on the integration of MNs with biomedical devices. Finally, we provide a survey on clinical applications of MNs and discuss opportunities and challenges for the translation and commercialization of MNs as biomedical devices.

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Key points

- Microneedles allow on-demand and intelligent control of drug delivery and biofluid extraction by integrating innovative biotechnologies, rational material engineering and structural designs.
- Microneedles can be delivered to specific targets and overcome barriers of diverse tissues beyond the skin, augmenting the therapeutic efficacy of treatments.
- Extraction, detection and real-time measurement of biomarkers by microneedle-based biosensors can provide personalized and precise health-care monitoring as an integral part of a closed-loop theranostics system.
- Scale-up, cost-effective and sterilized fabrication, consistent performance, and regulatory guidance should be considered before the translation of microneedle-based technologies.

Introduction

Microneedles (MNs) are needles of tens to thousands of micrometres in length that can mechanically break the superficial skin layer (stratum corneum) to improve drug permeation¹. MNs provide a less invasive transdermal route than hypodermic needle injections, offering a promising alternative especially for chronic disease treatment and paediatric use. MNs offer variability in material choices for manufacturing and modification, allowing dosage and spatial control in drug delivery². They can provide localized drug release to improve biodistribution and avoid the loss of drug efficacy caused by circulation in the blood, denaturation, clearance and tissue infiltration. MNs show great promise in treating various skin diseases (for example, acne, atopic dermatitis, psoriasis and skin cancers)^{3–7}, local diseases (for example, obesity)^{8,9} and targeting skin-resident immune cells using immunomodulation reagents^{10–12}. MN-mediated vaccinations show higher immunogenicity and dosage-sparing effects than intramuscular injections as clinically tested to deliver influenza and polio vaccines^{13–16}.

Additionally, MNs can be applied for biosensing. They can sample and detect health-related analytes from skin capillary blood and interstitial fluid (ISF). Unlike blood, the dermal ISF does not clot and contains physiologically relevant analytes, making it a promising source for continuous monitoring^{17,18}. MN-based biosensors can further be integrated with electronic sensors to monitor one or multiple biomarkers^{19–22}. For example, glucose biosensors have been explored to meet the demands of continuous glucose monitoring for patients with diabetes. Real-time and painless glucose monitoring is essential for precise planning of insulin administration²³.

Various MN-based biomedical platforms have emerged based on functional materials, nanomedicines, sensing technologies, and mechanical and electronic devices to monitor health conditions and to precisely and smartly control the delivery of therapeutics (Fig. 1). Although the original target tissue was the skin, MNs can now target other tissues and organs, including the eyes, oral cavity, heart, gastrointestinal tract (GIT) and tumours²⁴. Alza Corporation first patented the concept of miniature needles in the 1970s²⁵. MNs were first applied for transdermal drug delivery in 1998 (ref. 26), followed by glucose detection in 2000 (ref. 27). Since then, the research and development of MNs have accelerated, with involved academic and industrial partners

including Kindeva (3M), BD, Debiotech, Zosano Pharma, Verndari, NanoPass, Bioserentach, Sorrento Therapeutics, Micron Biomedical, Zenomics and Merck. Different MN-based platforms are being submitted for clinical evaluation, and some products have been approved as transdermal microinjection and blood collection systems such as MicronJet, TAP device and OneDraw kit.

In this Review, we provide an outline of MNs as biomedical devices covering their materials, structures and engineering approaches. We further discuss the latest developments in MN technologies and integrated MN devices for controlled therapeutic delivery, transdermal biosensing and closed-loop theranostics. Finally, we investigate the status of clinical trials and the potential for translation and commercialization.

MN engineering

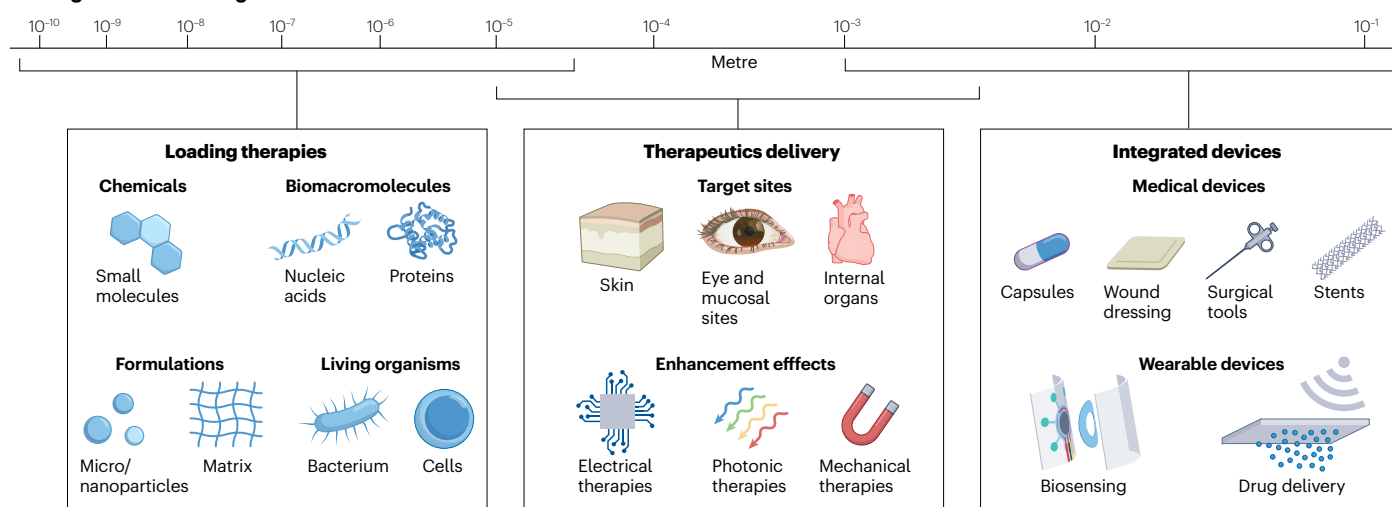
MNs are single needles or an array of minimized needles with dimensions from tens to thousands of micrometres that are attached to a baseplate to form patches. There are solid, hollow, porous, dissolving and swellable MNs (Table 1). Pressing solid MNs against tissues pokes the tissues and facilitates the permeability of subsequently applied drugs. Hollow and porous MNs allow the infusion of drugs in solutions into tissues. Dissolving and swellable MNs sustainably release the encapsulated drugs during the degradation and swelling of the MN matrix. MNs can perform simple functions; for example, metallic MNs are used for skin piercing, dissolving MNs for passive drug release and swellable MNs for sampling skin ISF. Their biomedical functions rely on the properties of the constructing materials (Table 1) and structural designs (Fig. 1b). It is critical to understand the capability and features of these parameters to improve the functions of MNs when developing advanced MN platforms (Fig. 1c).

Engineering parameters

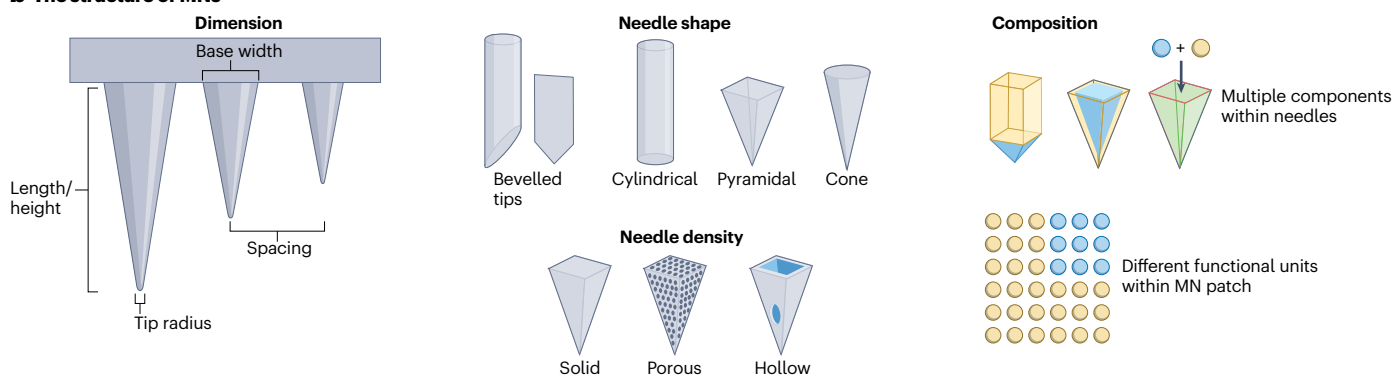
Materials. MNs can be made from silicon, metals, ceramics, glass, oligosaccharides and various polymers^{25,28,29} (Table 1). Although the first MN platforms, made from metals, have provided adequate mechanical properties for skin penetration, polymeric MNs are preferable owing to their ability to control drug release profiles and their biocompatibility³⁰. Polymers can be classified into non-degradable, biodegradable and water soluble. The chemical composition of polymers determines their water solubility, swellability or degradability in the hydrated skin environment. Other chemical and physical characteristics of polymers that can alter the mechanical and drug release properties of MNs include molecular weight, crosslinking degree, substitution degree, polydispersity, crystallinity and thermal transitions. For example, a silk MN patch can temporally control the release of multiple drugs². The MN patch is divided into three distinct regions of the baseplate, four outer MN arrays and one central MN array, in which the specific drugs are mixed with silk fibroin solution and precisely distributed through an inkjet printer. Subsequently, variable ethanol treatments are applied across the regions, resulting in uncrosslinked, partially crosslinked and fully crosslinked silk.

Structures. MNs should be rationally designed based on the application purposes, considering their structural features, which influence the mechanics and functions of MNs, and the capability to fabricate them. The structural features of MNs include geometrical parameters (for example, needle shape, needle surface and internal needle density), dimensional parameters (for example, tip radius, needle length, aspect ratio, needle-to-needle spacing, needle numbers per patch and

a Design of MN technologies at different scales



b The structure of MNs



c The functions of MNs

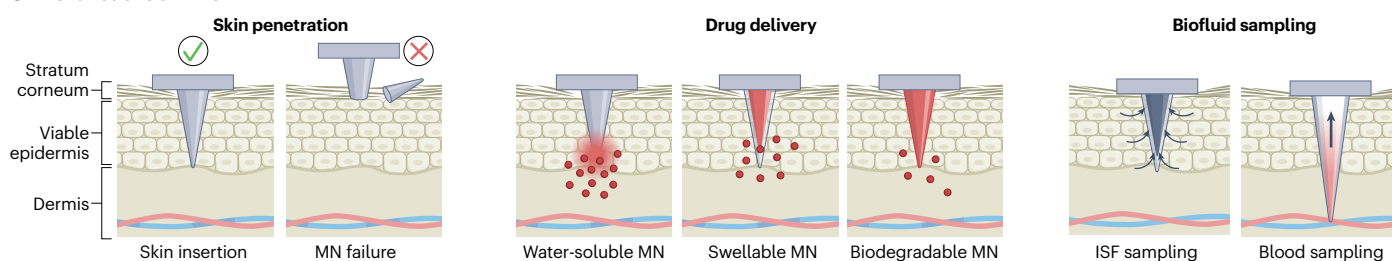


Fig. 1 | Design of MN-based technologies at different scales. a, Microneedles (MNs) can carry therapeutics of different sizes (from 10^{-10} to several 10^{-5} m), such as small molecules, biologics, engineered drug vehicles and living organisms, and control their release performance. The application of MNs extends from skin to various tissues, including hair, nail, eye, oral mucosa, heart and gastrointestinal tract. MNs have been designed with different sizes to interrupt biological barriers (barrier thickness ranging from 10^{-3} to 10^{-3} m), improving the targeted drug

delivery and therapeutic effects of electrical, photonic and mechanical therapies. MNs can be integrated with devices, such as capsules, stents and surgical tools, to deliver MNs to otherwise unreachable sites. Wearable MN devices can be designed with electronic units to monitor health care and manage disease under digital control. **b,** Structural parameters that should be considered in the design of MNs. **c,** Structure design to achieve biofunctions of MNs for skin penetration, drug delivery and biofluid sampling. ISF, interstitial fluid.

patch area) and composition (for example, the combination between hetero-materials) (Fig. 1b). Methods such as photolithography³¹, laser cutting³² and mechanical milling³³ are precise and digitally controlled to fabricate complicated MN designs. Alternatively, micromolding^{34,35}

and droplet-born air blowing³⁶ are more flexible methods for materials (especially polymeric materials) templating for fast fabrication of MNs. 3D printing techniques³⁷ (for example, stereolithography, digital light processing, continuous liquid interface production and two-photon

Table 1 | Representative MN materials and fabrication

Category	Materials	Selection considerations		Fabrication techniques	Applications	Refs.
		Advantages	Limitations			
Metal	Stainless steel, titanium, nickel, aluminium, palladium, copper and tungsten	Strong compressive and flexural strength; good electrical conductivity; easy sterilization and storage	Costly fabrication; biocompatibility concerns other than titanium and stainless steel	Reduction methods such as laser cutting, laser ablation and etchings; micromolding followed by sintering methods	Solid MNs for skin treatment, microelectrode; coated MNs for drug delivery; hollow MNs for microinjection; porous MNs for drug delivery and ISF sampling	84,197–200
Silicon	Silicon	Strong mechanical strength	Easy fracture; costly fabrication; biocompatibility concerns	Lithography and etching methods		26,31,201
Glass	Borosilicate glass	Strong mechanical strength	Easy fracture; manual fabrication; biocompatibility concerns	Pulling technique	Hollow MNs for microinjection and ISF sampling	202,203
Ceramics	Alumina, zirconia, calcium sulfate-based and calcium phosphate-based ceramics	Strong mechanical strength; tunable porosity; good biocompatibility	Easy fracture	Micromolding followed by sintering methods; 3D printing (photopolymerization)	Porous MNs for drug delivery and ISF sampling	204–208
Sugar glass	For example, maltose, sucrose, trehalose, mannitol, xylitol and galactose	Strong mechanical strength; good biocompatibility	Hygroscopic	Micromolding; drawing lithography	Dissolving MNs for drug delivery; solid MNs for pretreatment of skin	209,210
Non-degradable polymers	For example, PDMS, epoxy, photoresists, PE, PP, nylon, PMMA and PS	Good mechanical strength	Relatively low biocompatibility	Laser ablation; micromilling; 3D printing; micromolding (solvent casting, hot embossing)	Solid MNs and coated MNs for transdermal delivery	28
Biodegradable polymers	For example, PLA, PGA, PLGA, PCL, PC, and natural crosslinked proteins and polysaccharides	Excellent biocompatibility; tunable degradation and/or dissolution profiles; simple fabrication	Relatively low mechanical strength	Micromolding; 3D printing (fused deposition modelling)	Solid MNs for drug delivery with various release profiles	69,72,211
Water-soluble polymers	For example, polysaccharides, PVA, PVP, PAA, PEG, PMVE–MA, gelatin and nucleic acids			Micromolding; drawing lithography techniques	Dissolving MNs for drug delivery; swellable MNs for ISF sampling and biosensing	29,212,213

ISF, interstitial fluid; MN, microneedle; PAA, poly(acrylic acid); PC, polycarbonate; PCL, polycaprolactone; PDMS, polydimethylsiloxane; PE, polyethylene; PEG, polyethylene glycol; PGA, poly(glycolic acid); PP, polypropylene; PLA, polylactic acid; PLGA, poly(lactic-co-glycolic acid); PMMA, poly(methyl methacrylate); PMVE–MA, poly(methyl vinyl ether-co-maleic anhydride); PS, polystyrene; PVA, polyvinyl alcohol; PVP, polyvinylpyrrolidone.

polymerization) are an additive manufacturing technology to rapidly produce 3D prototype MNs with high accuracy and customizable shape and dimension. However, each technique may be limited by fabricating materials, equipment, production time and cost^{38–40}.

MN biofunction

Skin penetration. Skin penetration is the primary function of MNs, followed by drug delivery or biosensing. Skin is the primary barrier to protect the body, with the stratum corneum as the outermost layer. The stratum corneum comprises 15–20 layers of dead corneocytes with a thickness of approximately 10–40 µm, providing barrier functions against physical, chemical and biological exogens^{41,42}. Specifically, the stratum corneum has the stiffest mechanical strength (elastic modulus approximately 3.5–1,000 MPa)⁴³ and the highest electrical impedance (>105 Ω)⁴⁴ among all skin layers. The stratum corneum and the underneath viable epidermis construct the full avascular epidermis (approximately 100–150 µm thick). Unlike the epidermis, the dermis

and hypodermis underneath have abundant blood and lymphatic vessels, nerves, and skin appendages.

MNs should penetrate the stratum corneum or epidermis to deliver drugs into the viable skin (epidermis and dermis) to further diffuse to the systemic circulation. MNs should generate sufficient pressure to pierce the skin and avoid bending, fracture, or breaking during penetration^{45–48}. Structural designs and materials influence the mechanical behaviour of MNs. Materials with good compressive strength can support a porous or hollow structure, for example, porous ceramic MNs and hollow glass MNs. Flexural strength (the ability to withstand bending forces) of materials is also important. For example, silicon, ceramic and glass MNs could easily fracture during application. By contrast, metallic MNs are less prone to bending, and Dermal roller, made of stainless steel, can be used for repeated treatments. Combining materials with different mechanical properties can improve the performance of materials with low mechanical strength such as blending with another stiffer material to form a matrix or building a

strong core-shell structure. For example, a poly(lactic-co-glycolic acid) (PLGA) outer shell can protect MNs delivering a gelatin methacryloyl hydrogel and encapsulated mesenchymal stem cells into the skin to promote wound regeneration⁴⁹. Applicators can be designed to stretch the skin and perform faster insertion with higher pressing force than manual insertion. Thus, combining the functionality of the applicator into integrated MN devices could improve skin penetration^{50,51}.

Owing to the elasticity of the skin, MNs can detach from the skin by removing the insertion pressure⁵², which affects long-term applications for prolonged drug delivery and continuous health-care monitoring. Therefore, MNs are designed to maintain skin attachment after insertion using adhesive materials to improve interactions between MNs and the skin surface and tissues. Adhesive materials include acrylic adhesive⁵³ and polymers with adhesion properties (for example, mussel adhesive protein^{54,55}, polydopamine⁵⁶, silk fibroin methacryloyl⁵⁷, gelatin methacryloyl⁵⁸). In addition, structural designs can improve the skin adhesion of MNs such as microhooks^{59–61}, inclined arrays^{62,63}, and mechanical interlocking structures⁶⁴ and suction cups⁶⁵, which prevent detachment of MNs from the skin after application.

Drug delivery. MN patches can be designed to deliver sub-milligram amounts of drugs (large MN patches can deliver approximately 10 mg of drugs^{66,67}). Inserted MN tips can deliver more cargo than diffusion from the baseplate⁶⁸. Therefore, MNs are advantageous when delivering potent drugs (for example, insulin, growth hormone and vaccines) that take effect with a dose below 1 mg (ref. 69). For drugs with lower potency, MNs are positioned for precise localized delivery to reach an effective concentration in target tissues, and MN dimensions can be adjusted to achieve a high drug-loading capacity (for example, chemotherapeutics and monoclonal antibodies at the tumour site and steroids in the skin). To improve the loading of low-potency drugs, drugs can be loaded in the matrix of dissolving MNs. Thus, selecting and engineering matrix polymeric materials can improve drug loading to maximize drug doses in MNs^{3,68,70}. For example, MNs made from a matrix composed of β -cyclodextrin-modified gelatin methacryloyl enhance the loading capacity of the water-insoluble drug curcumin⁷⁰. The hydrophobic cavities of β -cyclodextrin can help solubilize hydrophobic drugs through intermolecular interactions. For example, with 40% β -cyclodextrin, the loading capacity of curcumin increases from 85 mg ml⁻¹ to 466 mg ml⁻¹, compared to the unmodified gelatin methacryloyl MNs⁷⁰.

MNs can also provide predefined release profiles of loaded drugs to achieve dosage control. Polymers can provide diverse release profiles, ranging from fast release in minutes to extended release over a few months. After inserting MNs in the skin, the encapsulated drugs are released from the matrix through passive diffusion, dissolution or degradation of polymers in contact with ISF. Programmable release of one or multiple drugs can be achieved through a suitable MN design. For example, core-shell MNs have achieved multiple and autonomous burst releases of vaccines in designed intervals from days to months⁷¹. Here, the poly(vinylpyrrolidone) core is loaded with Prevnar-13 (a clinically approved pneumococcal vaccine) and the shell is composed of biodegradable PLGA. Programmable release of payloads to perform prime and two-time booster vaccination on days 2, 10 and 40 can then be achieved by tailoring the degradation kinetics of the PLGA shells. These MNs were shown to protect rats, challenged with a lethal dosage of *Streptococcus pneumoniae* bacteria, against infection, with comparable results to subcutaneous injection at multiple time points⁷¹.

Biofluid sampling. Generating microchannels in the skin upon MN insertion allows sampling of skin ISF or blood for biomarker analysis for diagnosis^{28,72}. Taking a blood sample remains the gold standard in clinical practice. MN-based blood sampling aims to provide a less painful and self-administered workflow than using a hypodermic needle and syringe, allowing the collection of dozens to hundreds of microlitres of blood samples. As the skin vascular network is concentrated in the papillary dermis and deeper dermal-subcutaneous interface⁷³, hollow or solid metallic MNs can be designed with a length ranging from 1,000 μ m (ref. 74) to 1,800 μ m (refs. 75,76) to access and poke capillary vessels. MNs are then connected to a pre-sealed⁷⁷ or press-generated⁷⁸ vacuum chamber that triggers blood flow from capillaries and stores the samples in the device.

ISF is another informative source of biomarkers. ISF buffers between the micro-vasculature and local cells; thus, it contains analytes from blood and local biomarkers^{18,79}. Blood biomarkers enter ISF through transcapillary filtration, in which small molecules (for example, ions, glucose and alcohol) keep a similar concentration to that in the blood plasma but proteins, lipids and protein-binding hormones are substantially diluted (for example, ISF only contains approximately 20–25% of plasma lipoprotein)⁸⁰. ISF is abundantly present in the extracellular space in the viable epidermis and dermis. MNs for ISF extraction are commonly designed with a length of 100–1,000 μ m to access dermal ISF and avoid blood contamination. Even though ISF makes up approximately 40% of the skin volume, the flow of ISF is restrained by extracellular matrix components such as collagen and glycosaminoglycans. MN-based techniques can withdraw up to dozens of microlitres of ISF from the skin through three major mechanisms.

The first mechanism is to use solid MNs to poke the skin, followed by the removal of MNs and suction of ISF from the skin microchannels using a vacuum⁸¹ or liquid-absorbing paper⁸². The second is using hollow⁸³, porous^{84–86} and grooved solid MNs⁸⁷ to generate a capillary effect for ISF extraction. The capillary force is affected by factors such as lumen diameter, pore sizes and surface wettability. In addition, external pressure⁸⁸, vacuum⁸⁵ and electro-osmotic flow⁸⁹ are applied to increase the sampling speed of ISF. The third mechanism involves swellable hydrogel MNs that comprise many hydrophilic groups in the crosslinked polymeric network, which can absorb ISF through swelling. The extraction volume and speed of swelling can be affected by the water affinity of polymers (for example, crosslinked hyaluronic acid can swell and absorb water faster than poly(methyl vinyl ether-co-maleic anhydride)⁹⁰), crosslinking degrees⁹⁰ and osmotic pressure⁹¹. After ISF sampling by swellable MNs, post-processing procedures (for example, centrifugation against recovery buffer) are required to recover ISF from MNs into the liquid phase for subsequent analysis. For example, swellable MNs are made of polyvinyl alcohol (PVA) and chitosan, in which PVA can be thermally degraded⁹². With gentle heating at 60 °C for 15 min, the PVA-chitosan matrix and extracted biomarkers can be dissolved in water for further analysis. Four biomarkers, including glucose, chloride, lactate and bovine serum albumin, can be tested. However, bovine serum albumin has a lower thermal stability than the other biomarkers and requires lower heating temperature and duration for recovery⁹². Additionally, MNs can be modified with biorecognition molecules (for example, antibodies⁹³ and DNA aptamers⁹⁴) to concentrate analytes and avoid dilution of target biomarkers in post-processing. For example, the surface of porous trimethylolpropane triacrylate resin MNs can be modified with a DNA aptamer, which can specifically bind lipopolysaccharide for extraction and detection⁹⁵. In another design, an alginate hydrogel layer was coated on a solid poly-L-lactide MN⁹⁶. Ovalbumin

(as an antigen) and CpG (as an adjuvant) were incorporated in the outer hydrogel layer to recruit and sample leukocytes for analysis of the local immune cell population⁹⁶.

Integrated MN platforms

Advanced MN platforms can be designed for controlled drug release by constructing MNs with materials fabricated for responsive release. Additionally, MNs can be integrated with electronic devices to improve the convenience and delivery of therapeutics, biosensing and closed-loop theranostics.

MNs for therapeutics delivery

Pharmaceuticals delivery. During the delivery of proteins, nucleic acids, cells and nanomedicines, their therapeutic efficiency may be lost owing to drug administration, absorption and distribution prior to arrival at target sites. MN delivery platforms can be designed to improve the bioavailability of these drugs and provide spatially and/or temporally controlled delivery. MNs can provide localized delivery to target sites and minimize side effects, especially for drugs with considerable toxicity, such as anticancer drugs, to healthy tissues. For example, a bio-orthogonal catalytic MN patch can activate innocuous prodrugs in situ to active drugs at the tumour site⁹⁷. The MN patch is made of a PVA hydrogel containing Pd nanoparticles deposited onto TiO₂ nanosheets. Once exposed to the prodrug solution, the PVA matrix swells and embedded Pd catalysts can cleave the protective group for drug activation. This catalytic MN patch is locally applied on the skin surface at a melanoma tumour site in mice to locally transfer the systemically injected prodrug N-allyloxycarbonyl-caged doxorubicin into activated doxorubicin. Thus, the systemic toxicity of the metal catalysts and chemotherapy is minimized to achieve drug release with precise spatial control⁹⁷.

MN patches can be easily applied to superficial targets, such as skin, scalp, oral cavity and eye, but remain challenging to interface with internal sites in the body. However, MNs can be integrated with medical devices to reach inaccessible regions. For example, MNs integrated with a commercial balloon catheter can access and deliver drugs into blood vessels^{98,99}. The MN drug-eluting balloon (DEB) can deliver paclitaxel as a model drug¹⁰⁰; here, MNs on the surface of the DEB improve endovascular delivery and thus decrease the stenosis area compared to unmodified DEB in an atherosclerosis rabbit model¹⁰⁰.

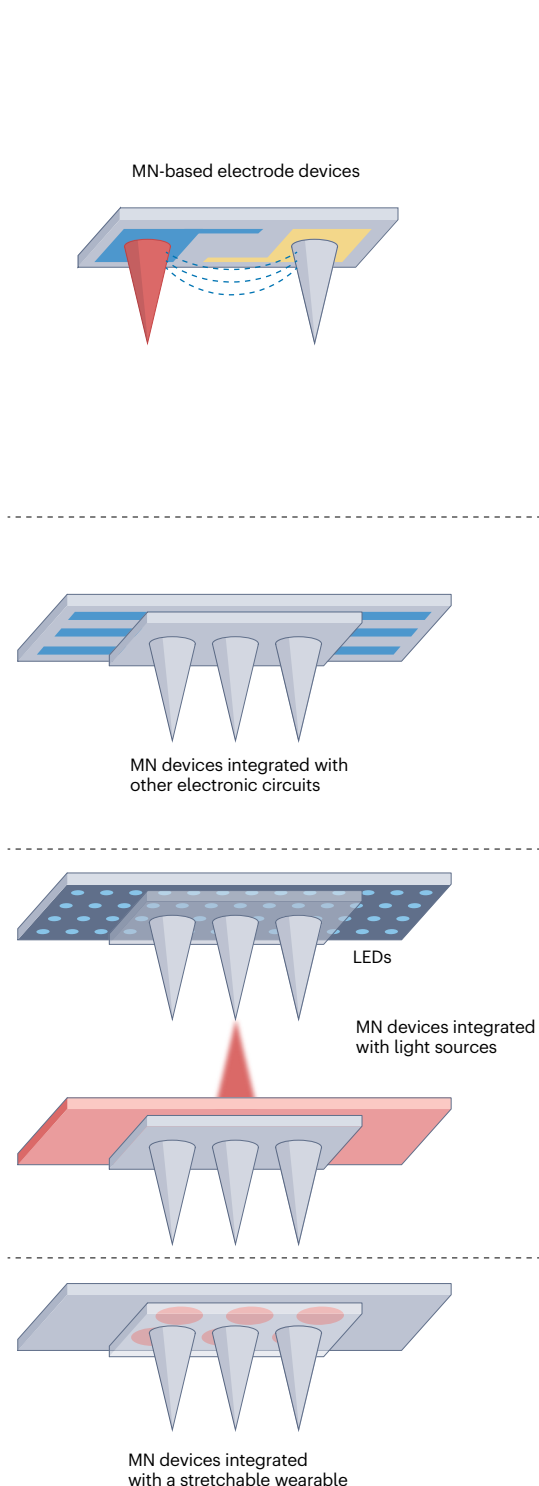
MN-integrated robotics enable oral administration to improve the absorption of macromolecules, such as insulin and mRNA vaccines in the GIT^{61,101–104}. MNs act as reservoirs and protectors of drugs and robotics control the trajectory and orientation to accurately insert MNs into GIT tissues. For example, poly(vinylpyrrolidone)-dissolving MNs are laminated to the flexible arms of a lumen-unfolding MN injector combined with a spring to form a capsule¹⁰². After oral administration, the polymers (poly(methacrylic acid-co-ethyl acrylate) and polyethylene glycol) supporting the spring inside the capsule dissolve to trigger release of the lumen-unfolding MN injector in the GIT at a pH of less than 5.5. The flexible arms then unfold to firmly press the MNs against the intestinal wall to penetrate the epithelial barrier, followed by the dissolution of MNs for insulin release¹⁰². In another example, a magneto-responsive MN robot made of polarized magnetic NdFeB particles allows manipulation of the movement of the MN patch with an external magnetic field¹⁰⁵. The MN robots are encapsulated in enteric capsules for oral administration and insulin release in the small intestine. A magnet is applied to adjust the position, orientation, and insertion of MNs into the intestine¹⁰⁵. MNs can also be delivered

remotely for first-aid purposes¹⁰⁶. For example, the unmanned aerial vehicle system can search and locate the emergency case and deliver a drug-loaded MN patch using a contact-triggered applicator. This platform can automatically deliver glucagon to rescue diabetic minipigs with hypoglycaemia symptoms¹⁰⁶.

MNs can be integrated with electronics for on-demand drug release to control drug delivery for chronic disease management (Fig. 2). For example, a stretch-responsive MN device has been developed with a drug depot microgel made from a stretchable commercial silicone elastomer (Dragon Skin 30) for on-demand release¹⁰⁷. Squeezing the microgel by applying pressure can initiate drug release. The MN device triggers insulin release by simple daily motion (for example, daily joint movements or intentional stretching) when tested on a diabetic mouse model¹⁰⁷. Such electronic MN drug-delivery systems generally feature three modules: the power system, an electronic circuit control element and a MN–drug reservoir unit in a miniature device such as small pieces of rigid printed circuit board¹⁰⁸ or stretchable electronics¹⁰⁹. Drug release can be triggered through various release mechanisms, for example, electrical control. MNs can be combined with miniaturized electronics, such as iontophoresis devices, for drug delivery¹¹⁰. For example, the electronic device combined with porous MNs can be composed of three units: a lithium battery-based power unit, a microprogrammed control unit chip, and a voltage-controlled constant current circuit to provide constant and adjustable iontophoresis current to control the drug diffusion rate through the MNs into the skin. Similarly, electric-responsive MNs can be designed with conductive polymers (for example, polyaniline, polyethylene dioxythiophene, poly(3,4-ethylenedioxythiophene) and polypyrrole) or conductive nano-architectures (for example, metallic nanoparticles and carbon-based nanomaterials)^{111,112}. Upon application of electric current, these materials can undergo reversible redox reactions that alter the charge of the materials and facilitate drug release^{4,113,114}.

Electronic devices can also activate drug release indirectly, for example, by transmitting electricity to heat, pressure and light to trigger drug release. For example, a microheater unit can be combined with MNs made from thermosensitive materials (for example, polycaprolactone and fatty acids) with a phase transition temperature slightly higher than body temperature^{115,116}. When the heating unit is activated, the MN–drug depot melts and releases the drug. An example of a pressure-activated system is an electronic device integrating hollow MNs, drug chambers and a micropump¹¹⁷. The micropump is controlled by a piezoelectric actuator that generates modulated pressure to regulate the flow rate of the liquid formulation from the drug chamber through the hollow MNs, delivering drugs into the skin¹¹⁷. Alternatively, a micro-light-emitting diode can be integrated with MNs to develop portable phototherapy devices. For example, MN-on-bioelectronics-equipped surgical forceps can guide ultraviolet (UV) light to deep tissues for photodynamic therapy (PDT)¹¹⁸. The high energy photon is generated from the light-emitting diode of a flexible bioelectronics device and delivered along MNs to the tissue. The device, which provides deeper penetration (3 mm) and higher energy (UV light of 365 nm) than near-infrared light in artificial brain tissue, has been applied on mouse subcutaneous glioblastoma and pituitary adenoma models in vivo. The dissolving hyaluronic acid MNs are placed directly onto the tumour tissues until dissolution to deliver nanoparticles containing a protoporphyrin-based PDT reagent, and the bioelectronic device provides UV photons that activate PDT. The mice were monitored for 3 weeks after treatment, and PDT treatments suppressed tumour growth and prolonged median survival times¹¹⁸.

a Integration of wearables with MNs



b Therapeutic functions of MN devices

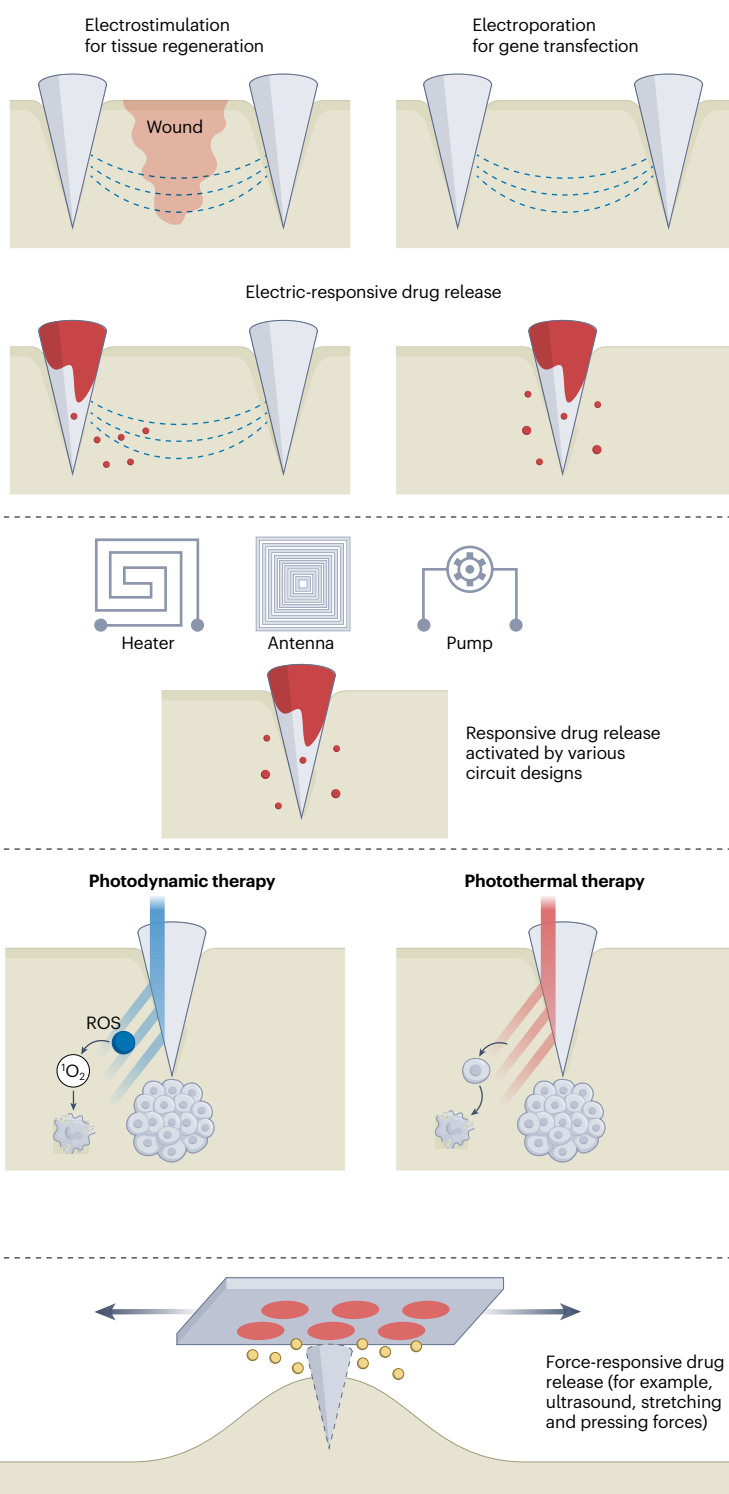


Fig. 2 | MN devices in the integration of wearables. **a**, Approaches to integrate microneedles (MNs) with electronic, photonic and tensile wearables. **b**, Wearable MN devices can achieve on-demand drug release through electrical, light, heat and mechanical triggers. With the interruption of biological barriers, wearable

MN devices enhance the penetration of physical stimuli and improve the effect of phototherapy and electrotherapy. LED, light-emitting diode; ROS, reactive oxygen species.

Physical stimuli-based therapy. Therapeutic modalities relying on physical stimuli, such as phototherapy and electrotherapy, can be combined with MNs. The skin barrier blocks or minimizes tissue penetration of some physical stimuli and reduces their efficiency. Thus, MNs can improve tissue penetration and precisely deliver the physical stimulus.

Phototherapy utilizes specific wavelengths of light for therapeutic purposes such as blue light to treat neonatal jaundice¹¹⁹. However, short-wavelength light sources, such as blue or UV light, show low tissue penetration with high energy. UV light can only reach a depth of less than 0.2 mm in the skin owing to scattering and absorption by surrounding tissues and fluids¹²⁰. MNs can be applied as light transmission channels to deeper tissue¹²¹. Indeed, MNs increase the skin penetration depth of blue light from approximately 0.2 mm to 2.5 mm, increasing total transmission by 8.9-fold. These waveguide MNs should be transparent and can be manufactured from polylactic acid and PLGA^{121–123}. MNs also permit the integration of photothermal or photosensitive nanomaterials, which convert light into heat (photothermal therapy) or reactive oxygen species (PDT) for cell damage and tumour destruction^{7,124}. For example, a MN patch for photothermal therapy, loaded with melanin, whole-tumour lysates and adjuvant, improves cancer immunotherapy¹²⁵. Upon near-infrared irradiation, melanin converts light to localized heat and releases cytokines that promote the recruitment and activation of immune cells. With MN vaccination, 87% of mice survived with complete melanoma rejection in a prophylactic mouse model, showing that the strategy has antitumour effects against primary and distant tumours¹²⁵.

Electrical stimulation can treat conditions such as wounds, obesity and hair loss as electrical impulses can influence cellular behaviour and tissue regeneration¹²⁶. MNs create a pathway across the high-impedance stratum corneum, reducing skin impedance, allowing higher current strength and facilitating accelerated regeneration. For example, a MN device with a self-powered triboelectric nanogenerator enables transcutaneous electrical stimulation of the cellular expression of growth factor receptors, preventing the desensitization of the epidermal growth factor pathway and promoting wound healing¹²⁷. MN-assisted electrical stimulation shows approximately 36% faster healing of full-thickness skin excision in mice with more neovessels and hair follicles, compared with electrical stimulation treatment alone. Another example is an implantable MN-based device for the delivery of anti-inflammatory drugs and periodic electrostimulation for muscle regeneration¹⁰⁹. This electronic device comprises a wireless power transmission electronic system (for example, magnesium electrical pattern on PLGA substrate) and a drug-loaded (for example, aspirin and ibuprofen) PLGA MN patch; the device is fully bioresorbable and can be remotely controlled to minimize surgical procedures¹⁰⁹.

By combining electroporation with MNs, the strength of the electrical field can be concentrated at the tips of the MN electrode to disrupt the cell membrane for nucleic acid transfection. Furthermore, MNs concentrate the electric field in the epidermis, which is particularly rich in antigen-presenting cells, to improve vaccination efficiency^{128,129}. For example, a MN electrode array electroporator (ePatch) consisting of a piezoelectric pulse generator and an array of metal MNs for DNA vaccination has been developed¹¹. The piezoelectric pulser is derived from the common household lighter to generate an electrical pulse with a thumb press. The MN electrodes are closely spaced to obtain sufficient electrical field strength for local electroporation. After intradermal injection of DNA plasmid for the SARS-CoV-2 spike protein, the ePatch was applied for electroporation in a mouse model. The DNA vaccine delivered by intradermal injection and the ePatch are well tolerated and

induce strong antibody responses against SARS-CoV-2, showing a ten-fold dose-sparing effect compared to intramuscular and intradermal injection without electroporation¹¹.

MN devices for sensing

For biosensing purposes, MNs can be designed to convert biological into readable signals. Combined with colourimetric, fluorometric or electrochemical assays, MNs can quantitatively and qualitatively analyse various biomarkers, including electrolytes¹³⁰, metabolites (for example, glucose²⁰, alcohol¹³¹, lactate¹³², uric acid and cholesterol¹³³), drug molecules (for example, vancomycin¹³⁴, tobramycin¹⁹, morphine and fentanyl¹³⁵), nucleic acids^{136–138}, proteins (for example, tyrosinase¹³⁹ and cytokines^{140,141}), exosomes⁹³ and cells⁹⁶. The analysis of target biomarkers can be in situ on the MN–skin interface or ex situ after sampling biofluids by MNs. MN-integrated electronic devices (MN electronics) utilize the functionalities of electronic components to miniaturize laboratory analysis into point-of-care testing.

Colourimetric detection. The colourimetric assay presents a visual colour change, reflecting the levels of target biomarkers. Redox-responsive chromogenic dyes and nanomaterials are two types of colour indicators to develop concentration-dependent colour changes in MN sensors. Reagents for colourimetric assays (for example, enzymes, chromogenic dyes and nanoparticles) can be either encapsulated in MN structures^{142–144} or embedded in paper strips attached to MNs^{145–149}. Dermal tattoo MN biosensors made of dissolving hyaluronic acid, in which different chromogenic dyes develop distinguishing colour changes for each analyte, can be used to detect multiple analytes¹⁵⁰. For the uric acid assay, uric acid reacts with uricase to produce H₂O₂, which works with peroxidase to oxidize 3,3',5,5'-tetramethylbenzidine dye, developing a blue colour. Similar mechanisms are applied for other analytes, for example, glucose, pH, lactate and cholesterol¹⁵⁰.

Nanostructured materials can induce colour changes under aggregation, structural rearrangement and morphology transition¹⁵¹. Colloid gold nanoparticle-based lateral flow immunoassays are commonly used for fast colourimetric detection of bioanalytes. The paper-based assays can be connected to hollow SU-8 (an epoxy-based negative photoresist) MNs¹⁴⁸, porous polylactic acid MNs¹⁴⁹ and poly(methyl vinyl ether-co-maleic anhydride)-based swellable MNs¹⁵² for fluid-sampling and to screen infectious diseases. Alternatively, colloidal crystals (periodically arranged silicon dioxide nanoparticles) are chromogenic components in colourimetric MN devices¹⁴⁴. A redshift or blueshift in the reflection spectrum of colloidal crystals indicates concentration changes in analytes. For example, to detect glucose, colloidal crystals are embedded within a polyacrylamide hydrogel that is modified with glucose-responsive fluorophenylboronic acid. When glucose levels increase, the glucose molecules interact with the boronic acid groups in the matrix, increasing its hydrophilicity, leading to swelling of the hydrogel and expansion of the distance between the silicon dioxide nanoparticles. Consequently, the colloidal crystals undergo a noticeable redshift in colour, transitioning from violet to blue and, ultimately, to green¹⁴⁴.

Fluorometric detection. The fluorescent immunoassay is also achievable by MN devices for sensitive analysis of proteins and nucleic acids. MNs selectively capture biomarkers in biofluids, followed by the ex situ addition of fluorescent secondary antibodies and aptamer probes to fluorometrically quantify biomarkers bound to the MNs^{140,153}. For example, photonic crystal (PhC) barcode-MNs can detect multiple

biomarkers simultaneously¹⁵⁴; here, a PhC barcode is made with silicon dioxide nanoparticle-based colloidal crystal microbeads. The PhC barcode is immobilized with specific recognition probes for each biomarker and integrated into the poly(ethylene glycol) diacrylate-based hydrogel MN patch. Upon MN insertion in the skin, biomarkers in ISF are concentrated on the barcodes through binding with the recognition probe. The fluorescent probes are then added *ex situ* to form sandwich immunocomplexes so that the specific biomarker can be distinguished and quantified with different colour coding and fluorescent intensity. PhC barcode-MNs can detect three inflammatory cytokines (tumour necrosis factor (TNF), IL-1 β and IL-6) at concentrations ranging from 10 to 500 ng ml⁻¹ in solution, showing similar accuracy to histological analysis and blood testing in mice with sepsis¹⁵⁴. MNs can also be modified with complementary nucleic acid strands for sequence-specific detection of nucleic acid biomarkers through complementary base pairing. Capturing the target sequence can be fluorescently detected by DNA intercalator dyes such as SYBR Safe (a cyanine dye used as a nucleic acid stain)¹⁵³. Hydrogel MNs can also be modified with nucleic acid aptamers to detect various small molecules (such as glucose, uric acid and ATP) and proteins (such as thrombin and insulin). The fluorophore-conjugated DNA aptamers are quenched to show no fluorescent signal. Once the aptamers recognize and bind to the target biomarkers, the pre-quenched aptamers undergo structural changes to depart the quencher and generate fluorescent signals reflecting the presence of the targets^{155,156}.

Electrochemical detection. The electrochemical assay is a precise method for the quantification of analytes. MN devices can be loaded with enzymes (for example, glucose oxidase and lactase oxidase) for enzyme-based electrochemical detection¹⁵⁷. For example, a MN glucose biosensor based on a three-conductive electrode system has been developed using resin MNs coated with gold¹⁵⁸. The biosensor contains nine columns of MNs, of which groups of three are combined as the working, reference and counter electrodes. Glucose is first specifically oxidized by the immobilized glucose oxidase on the surface of the working electrode. The electron generated from the oxidation process is then presented to the working electrode through a redox mediator layer (Prussian blue), in which the electrical current reflects the glucose concentration.

Enzymatic electrochemical biosensors can detect multiple biomarkers (for example, glucose, alcohol, lactate and cholesterol). Different MN working electrodes can be assembled on the same patch to detect various biomarkers simultaneously. The potential crosstalk between different analytes should be considered and minimized in the multiplexed sensing mode. For example, crosstalk can appear if a locally generated middle product, such as H₂O₂, produced by the oxidation of glucose, diffuses from one working electrode to another, interrupting the accurate detection of other analytes. Glucose oxidase can be replaced with H₂O₂-insensitive flavin-dependent glucose dehydrogenase¹⁵⁹, which, when combined with a lactate oxidase-based biosensor, can detect lactate and glucose simultaneously without crosstalk. Another fully integrated MN sensor system can simultaneously detect multiple biomarkers (glucose and lactate, glucose and alcohol) with minimized crosstalk through the arrangement of two sensing regions with sufficient distance²¹. Non-enzymatic catalytic materials^{160–162} and affinity-based sensing^{19,22,163} are also suitable for MN electronic biosensing devices. For example, DNA aptamers can be used as recognition elements in the MN sensor to track the pharmacokinetics of tobramycin and vancomycin for precise dosing control¹⁹.

As another example, a nitric oxide sensor has been developed using conductive MNs functionalized with hemin, which allows specific binding with nitric oxide and electrochemically reduces it to generate electrical signals¹⁶¹.

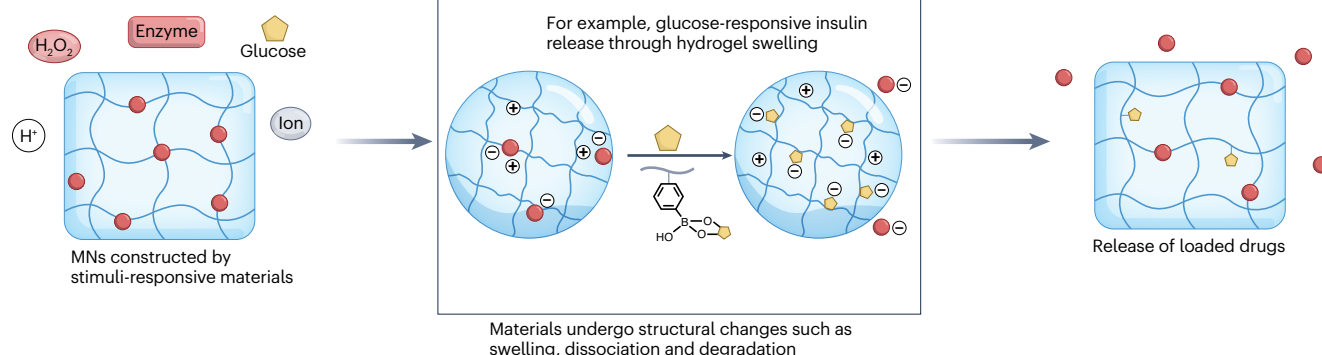
Electrochemical assays can also be integrated with wearable electronics. For example, a fully integrated MN sensor system consists of an array of multiple enzyme-based electrochemical-sensing probes, custom electronics and a smartphone application for data acquisition and visualization²¹. The system is designed for the simultaneous detection of glucose, lactate and alcohol. The custom electronics comprise various functional units, including a multichannel potentiostat, a microcontroller, a data transmitter and a power or charging system. Electrochemical analytical circuits control the potential of working electrodes and measure the currents and potentials. The microcontroller can set the parameters of electrochemical tests (for example, sensor channels, input voltage and running duration) and process the data from the tests (for example, built-in analogue-to-digital converter to generate digitalized data). The processed data is then wirelessly transmitted to a smartphone for visualization and analysis²¹. Wearable electronics (for example, smartwatches and stretchable electronics) are valuable for health monitoring, such as pulsatile blood flow, body temperature and exercise movements, using different sensory units such as thermal, tactile and optical sensors¹⁶⁴. MN-based transdermal biosensing technologies are noninvasive biochemical sensory units that can be integrated into wearable electronics for personalized health care²¹.

Closed-loop theranostic devices

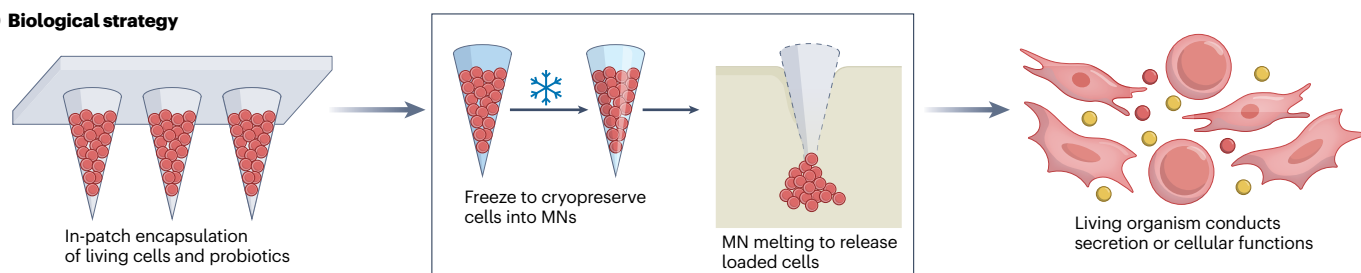
Theranostic systems combining sensing and on-demand drug release are essential for accurate drug regimens. For example, intelligent insulin delivery is expected to impact the management of diabetes^{165,166}. Closed-loop theranostics can be achieved through material, biological and computational strategies (Fig. 3).

Stimuli-responsive materials. Glucose-responsive MN patches can regulate blood glucose levels based on the reversible binding between glucose and phenylboronic acid¹⁶⁷. For example, MNs can be copolymerized from monomers, including cationic acrylates (such as 2-aminoethyl methacrylate hydrochloride and (dimethylamino) ethyl acrylate), N-vinylpyrrolidone and 3-(acrylamido) phenylboronic acid^{168,169}. The hydrophilic polymers and phenylboronic acid exhibit reversible swelling volume and charge changes in response to variable glucose levels, enabling the glucose-dependent release of antidiabetic drugs^{168,169}. Dual hormones of glucagon and insulin are introduced together in the responsive system to release insulin under hyperglycaemia and glucagon under hypoglycaemia⁶⁶. In the case of high blood glucose, the formation of a negatively charged glucose-borate complex causes the rapid release of negatively charged insulin but not of positively charged glucagon. Conversely, under hypoglycaemia, the charge of the polymeric matrix shifts from negative to positive, which inhibits the release of insulin and facilitates the release of glucagon. Using dual hormones mimics the physiological regulation of blood glucose and shortened the period of hypoglycaemia in a type 1 diabetic minipig model⁶⁶. The design of stimuli-responsive MN platforms requires consideration of the specific pathological conditions and selection of appropriate responsive motifs^{170–172}. Specific conditions, such as acidic conditions (for example, in the tumour microenvironment)¹⁷³, high levels of reactive oxygen species (for example, in inflammatory disorders)⁴ or the presence of specific enzymatic biomarkers (for example,

a Material strategy



b Biological strategy



c Digitalization strategy

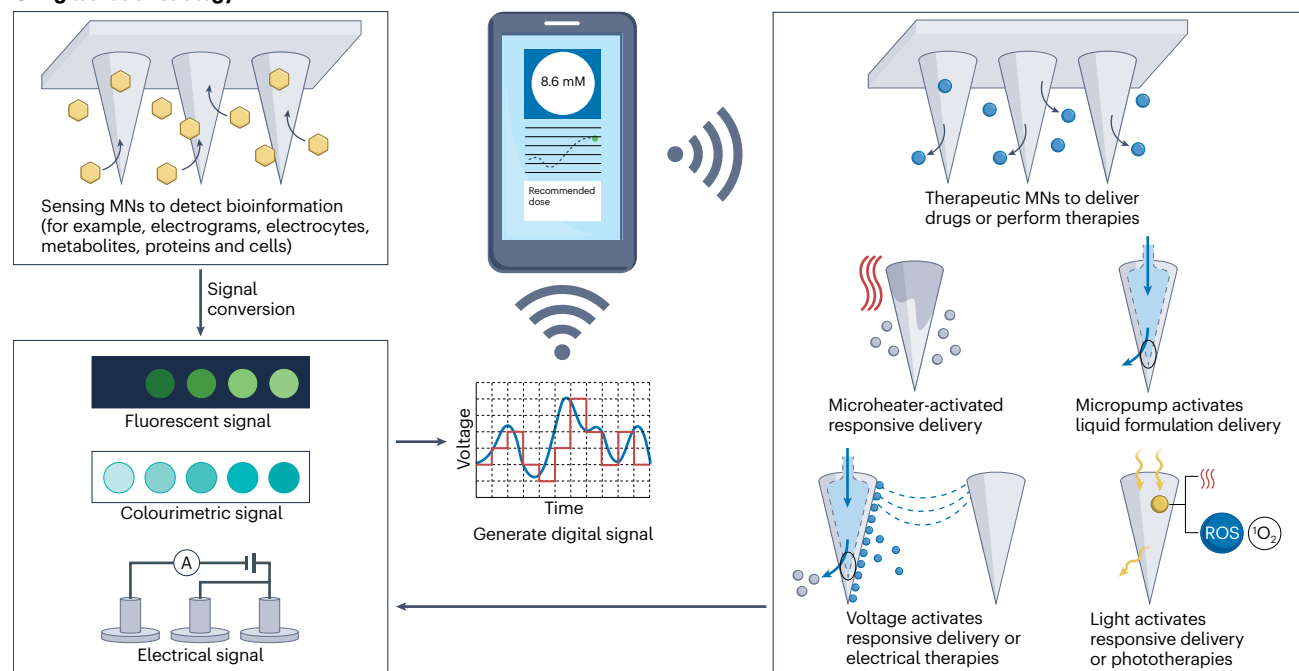


Fig. 3 | Different strategies to develop MNs for closed-loop theranostics.

a, Microneedles (MNs) can be constructed with responsive materials. The materials are rationally designed carrier systems to release drugs under physiological triggers; for example, insulin release under hyperglycaemic conditions. **b**, MNs can be loaded with functional living organisms. The living organisms are transplanted into target tissue using MNs to secrete desirable therapeutics in response to the surrounding environment; for example, living pancreatic β -cells for dynamic insulin secretion. **c**, MN electronic devices achieve closed-loop theranostics through digital computation and control. Sensing MNs

detect bioinformation from skin interstitial fluid; here, biosignals are recognized and converted to optical or electrical signals. The converted signals are further amplified, digitalized and transmitted to end-users (for example, smartphones) by integrated electronic units. The calculation, analysis, display of data and other functions, such as medication suggestion, medical history record and online medical consultations, can be achieved through the algorithm. When the user confirms the medication plan, therapeutic MNs can perform on-demand therapies or deliver drugs. ROS, reactive oxygen species.

Table 2 | Representative clinical trials of MN products in phase II and later stages from 2012 to 2022

Clinical trial number	Study title	Sponsor	Application(s)	Phase, status	MN device	Year
Vaccination						
NCT01707602	Routes of immunization and flu immune responses	Hôpitaux de Paris	Influenza vaccination	Phase I and phase II, completed	INTANZA 15; Vaxigrip; procedure: MN-based intradermal, transcutaneous, intramuscular injection	2012
NCT01813604	Immunogenicity of inactivated and live polio vaccines	Centers for Disease Control and Prevention	Polio vaccination	Phase III, completed	NanoPass device; MicronJet 600 MNs; polio vaccine	2014
NCT01686503	Intradermal versus intramuscular polio vaccine booster in HIV-infected subjects (IDIPV)	Eastern Virginia Medical School	Polio vaccination	Phase II, completed	NanoPass device; MicronJet 600 MNs; polio vaccine	2015
NCT02621112	HBV vaccine in patients with renal failure	The University of Hong Kong	HBV vaccination	Phase II and phase III, completed	HBV vaccine; Imiquimod Procedure: subcutaneous injection and MN-based intradermal injection	2019
NCT02329457	VZV vaccine for hematopoietic stem cell transplantation	The University of Hong Kong	VZV vaccination	Phase II and phase III, completed	Zostavax (VZV vaccine); intramuscular injection and MN-based intradermal injection	2019
NCT04394689	Measles and rubella vaccine microneedle patch phase 1-2 age de-escalation trial	Micron Biomedical	Measles-rubella vaccination	Phase I and phase II, recruiting	Dissolving MN patch; measles-rubella vaccine	2021
NCT05315362	Establishing immunogenicity and safety of needle-free intradermal delivery of mRNA COVID-19 vaccine	Leiden University Medical Center	COVID-19 vaccination	Phase II, recruiting	Solid MNs; mRNA-1273 vaccine	2022
Delivery of biologics						
NCT00837512	Insulin delivery using microneedles in type 1 diabetes	Emory University Thrasher Research Fund	Type 1 diabetes mellitus	Phase II and phase III, completed	900 µm hollow MNs; 9 mm subcutaneous insulin catheter; lispro insulin	2013
NCT05329532	Modi-1 in breast, head and neck, ovarian, or renal cancer (ModiFY)	Scancell	Triple-negative breast cancer; renal cell cancer; high-grade ovarian serous adenocarcinoma; head and neck squamous cell carcinoma	Phase I and phase II, recruiting	NanoPass device, MicronJet 600 MNs; Modi-1/Modi-1v; pembrolizumab	2022
NCT01674621	Phase 2 study of BA058 (abaloparatide) transdermal delivery in postmenopausal women with osteoporosis	Radius Health	Postmenopausal osteoporosis	Phase II, completed	Abaloparatide transdermal MN active patch	2020
NCT04064411	Efficacy and safety of abaloparatide-solid microstructured transdermal system in postmenopausal women with osteoporosis	Radius Health	Postmenopausal osteoporosis	Phase III, completed	Abaloparatide-solid microstructured transdermal system; abaloparatide prefilled pen for subcutaneous injection	2021
NCT03607903	Adalimumab microneedles in healthy volunteers	Centre for Human Drug Research, Netherlands	Local anaesthesia	Phase I and phase II, completed	NanoPass device; MicronJet 600 MNs; adalimumab	2018
NCT03203447	Suprachoroidal injection of triamcinolone acetonide with IVT anti-VEGF in subjects with macular oedema following RVO (TOPAZ)	Clearside Biomedical	Macular oedema retinal vein occlusion	Phase III, completed	Suprachoroidal microinjector; CLS-TA + Lucentis or Avastin (intravitreal anti-VEGF agent)	2021
NCT02949024	Suprachoroidal injection of CLS-TA alone or with aflibercept in subjects with diabetic macular oedema	Clearside Biomedical	Diabetic macular oedema	Phase I and phase II, completed	Suprachoroidal microinjector; aflibercept; CLS-TA	2021

Table 2 (continued) | Representative clinical trials of MN products in phase II and later stages from 2012 to 2022

Clinical trial number	Study title	Sponsor	Application(s)	Phase, status	MN device	Year
Delivery of biologics (continued)						
NCT03126786	Suprachoroidal CLS-TA with intravitreal aflibercept versus aflibercept alone in subject with diabetic macular oedema	Clearside Biomedical	Diabetic macular oedema	Phase II, completed	Suprachoroidal microinjector; aflibercept + CLS-TA; aflibercept + sham	2021
Drug delivery						
NCT02497846	TEOSYAL PureSense Redensity (I) injection using MicronJet needle in the treatment of crow's feet wrinkles	Teoxane	Crow feet wrinkles	Phase IV, completed	TEOSYAL PureSense Redensity (I)/MicronJet	2016
NCT02632110	Microneedle lesion preparation prior to aminolevulinic acid photodynamic therapy (ALA-PDT) for AK on face	DUSA Pharmaceuticals	Actinic keratosis	Phase II, completed	MNs; 20% ALA topical solution; investigational blue light	2017
NCT03012191	Gentamicin for RDEB	University of Southern California	RDEB	Phase I and phase II, completed	MN roller; gentamicin sulfate	2019
NCT03097315	Suprachoroidal injection of CLS-TA in patients with non-infectious uveitis	Clearside Biomedical	Non-infectious uveitis	Phase III, completed	Suprachoroidal microinjector; CLS-TA	2021
NCT04928222	Placebo microneedles in healthy volunteers (part I) and efficacy/safety of doxorubicin microneedles in basal cell cancer subjects (part II)	SkinJect	Basal cell carcinoma	Phase I and phase II, recruiting	Doxorubicin-loaded dissolving MNs; placebo-containing MNs	2022
NCT02745392	Safety and efficacy of zp-zolmitriptan intracutaneous microneedle systems for the acute treatment of migraine	Zosano Pharma	Acute migraine	Phase II and phase III, completed	Intracutaneous MN system; zolmitriptan and placebo	2018
NCT03282227	A study to evaluate the long-term safety of M207 in the acute treatment of migraine	Zosano Pharma	Migraine	Phase III, completed	Intracutaneous MNs; zolmitriptan	2020
NCT04066023	Double-blind comparison of the efficacy and safety of C213 to placebo for the acute treatment of cluster headaches	Zosano Pharma	Cluster headache	Phase II and phase III, completed	Zolmitriptan-coated titanium MNs; placebo	2022
NCT05078463	Efficacy of transdermal microneedle patch for topical anesthesia enhancement in paediatric patients with thalassemia	Universiti Kebangsaan Malaysia Medical Centre	Local anaesthesia	Phase II, completed	Maltose MNs; PVA-PET MNs; EMLA topical cream (lidocaine 2.5% and prilocaine 2.5%)	2022
NCT01789320	Safety study of suprachoroidal triamcinolone acetonide using microneedle to treat uveitis	Clearside Biomedical	Non-infectious uveitis	Phase I and phase II, completed	Suprachoroidal microinjector; Triesence (triamcinolone acetonide)	2021
NCT02255032	Suprachoroidal injection of triamcinolone acetonide in subjects with macular oedema following non-infectious uveitis	Clearside Biomedical	Non-infectious uveitis	Phase II, completed	Suprachoroidal microinjector; CLS-TA	2021

Table 2 (continued) | Representative clinical trials of MN products in phase II and later stages from 2012 to 2022

Clinical trial number	Study title	Sponsor	Application(s)	Phase, status	MN device	Year
Drug delivery (continued)						
NCT02595398	Suprachoroidal injection of CLS-TA in subjects with macular oedema associated with non-infectious uveitis	Clearside Biomedical	Non-infectious uveitis	Phase III, completed	Suprachoroidal microinjector; CLS-TA	2021
Diagnosis						
NCT02712593	A study investigating the effects of Niagen in healthy adults	KGK Science	Muscle metabolite sampling	Phase II, completed	MNs for collecting metabolites from muscle tissue	2018

Data were retrieved from the National Institute of Health ClinicalTrials.org database. ALA, aminolevulinic acid; CLS-TA, triamcinolone acetonide injectable suspension; EMLA, eutectic mixture of local anaesthetics; HBV, hepatitis B virus; MN, microneedle; PET, polyethylene terephthalate; PVA, polyvinyl alcohol; RDEB, recessive dystrophic epidermolysis bullosa; VEGF, vascular endothelial growth factor; VZV, varicella zoster virus.

thrombin in thrombosis)¹⁷⁴, can be exploited as stimuli for responsive MN systems. Additionally, nucleic acid aptamers targeting nucleic acids, drugs, metabolites and proteins are widely applied in target enrichment and detection. Nucleic acid-based structures (for example, DNA origami^{175–177}, genetic circuits^{178–180} and DNA hydrogels¹⁸¹) are programmable to regulate on-demand release of therapeutics for a closed-loop system. For example, MNs can deliver responsive nucleic acid nanodrugs to treat rheumatoid arthritis¹⁸². The nanodrugs are assembled by three DNA aptamers targeting two therapeutic targets (protein DEK and TNF receptor 1) and later disassemble in the presence of ATP to release therapeutic DNA strands. Thus, the nanodrugs show an ATP-responsive anti-inflammatory effect with dual targets¹⁸².

Living cell systems. Engineered mammalian cells and bacteria can act as living therapeutics that sense the biological environment and respond through cellular functions. Such living therapeutics can be delivered by MNs, which can then be retrieved to improve the biosafety of genetically engineered organisms and cells and reduce immune rejection. For example, exogenous pancreatic β -cells can be encapsulated in MN patches for glucose-responsive insulin delivery¹⁸³. The live pancreatic β -cells are encapsulated in alginate microgels and embedded in a hyaluronic acid hydrogel, which is externally located on the baseplate of the MN patch. Owing to the limited diffusion of glucose from the skin through the MN patch, pancreatic β -cells produce insufficient insulin responding to the hyperglycaemic state. To trigger the cellular secretion of insulin, synthetic glucose amplifier nanoparticles and α -amylose are loaded in the MN tips, which produce glucose within the MNs in a glucose-responsive manner. The glucose amplifier nanoparticles are self-assembled by hypoxia-sensitive hyaluronic acid (modified with the 2-nitroimidazole group), which comprises glucose oxidase, α -amylose and glucoamylase. The glucose amplifier nanoparticles disassociate under high levels of blood glucose and the presence of glucose oxidase, which releases α -amylose and glucoamylase to hydrolyse α -amylose. The system shows blood glucose regulation for up to 10 h within the normal physiological range in type 1 diabetic mice¹⁸³.

Various types of somatic cells (for example, cardiac cells^{184,185}, keratocytes and melanocytes¹⁸⁶, and stem cells^{49,187}) can be locally transplanted using hollow, solid and porous MNs to promote tissue regeneration. MNs are also used to deliver living probiotics, such as *Lactobacillus*¹⁸⁸, *Bdellovibrio bacteriovorus*¹⁸⁹ and *Bacillus subtilis*¹⁹⁰, to treat skin and ocular infections. However, maintaining the viability

of fragile living cells in a dehydrated state is challenging. Thus, hybrid core–shell structure MNs have been developed to encapsulate mesenchymal stem cells in a hydrogel core to preserve cell viability⁴⁹. Another strategy is loading engineered therapeutic cells, such as chimaeric antigen receptor T cells, in porous MNs¹⁹¹. Loading of the cells in the pores of the PLGA MN tips allows their delivery in the tumour bed in a dispersed manner and preserves their activity¹⁹¹. Proliferating cells cannot survive long in MNs of limited nutrition supply. Cryogenic MNs have been developed containing a cryogenic medium with pre-suspended cells to maintain cell viability after 1 month of storage in liquid nitrogen¹⁹². Further studies on the fate of delivered cells, host reactions, cell types and indications are required to improve the performance of living cells in combination with MNs to reduce storage restrictions and improve cell bioactivities.

Electronic devices. MN-integrated electronics combining on-demand drug release and sensing systems can provide minimally invasive portable devices for personalized medicine¹⁹³. For example, a fully integrated theranostic device for closed-loop management of diabetes includes an electrochemical sensor for continuous monitoring of blood glucose and an electro-osmotic pump for insulin delivery. The biosensor and electro-osmotic pump are controlled by the printed circuit board, which integrates the glucose signal sensor, amperometric electrochemical analyser, signal processing, control and feedback, and iontophoretic drug delivery. The high glucose signal generates a specific electric current, which turns on the electro-osmotic pump through signal transmissions from the circuit board¹⁹⁴.

Furthermore, MN-integrated theranostic devices for the management of other diseases have been proposed; for example, one or more disease-related key biomarkers can be continuously monitored using a MN electronic sensing system to digitalize, transmit and analyse biomarker levels and calculate required dosages for on-demand drug release⁴². Drug monitoring should be considered during the application of MN systems to ensure the safety and efficiency of therapeutics and continuously optimize the personalized algorithm. Physicians can further remotely review the generated health and drug administration data with a guarantee of personal privacy.

Clinical trials and translation

Many pharmaceutical companies are engaged in the development and manufacturing of MN technologies. Through the collaboration of pharmaceutical companies with universities and research institutes,

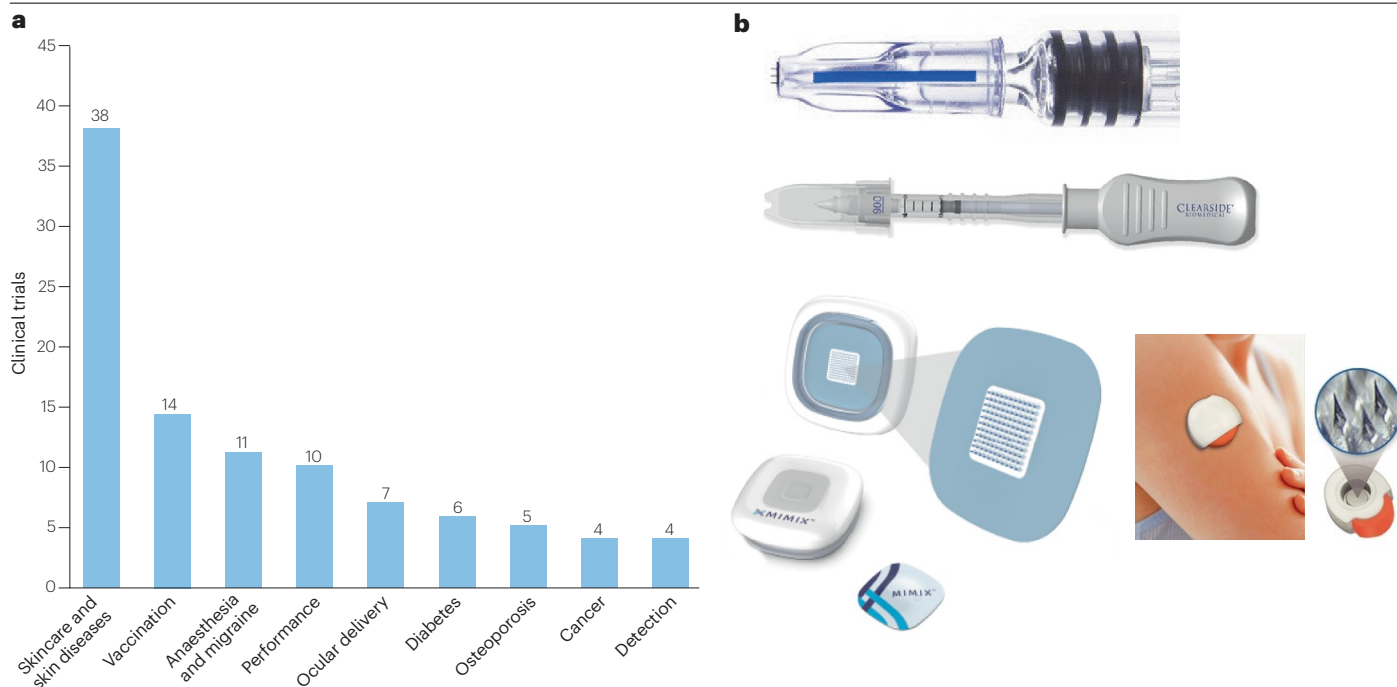


Fig. 4 | Clinical translation of MNs. **a**, Applications of microneedles (MNs) in clinical trials. Data were obtained by analysing 138 clinical studies conducted between 2007 and 2022 from the [National Institute of Health ClinicalTrials.org](https://www.clinicaltrials.gov/) database by searching the keyword ‘microneedle’. **b**, Representative pharmaceutical MN products, including MicroJet 600 (NanoPass) (top),

SCS Microinjector (Clearside Biomedical) (middle), MIMIX Patch (Vaxess) (bottom left) and MicroCor (Corium) (bottom right). Part **b**, top, image courtesy of NanoPass Technologies Ltd. Part **b**, middle, image courtesy of Clearside Biomedical. Part **b**, bottom left, image courtesy of Vaxess and Gerd Schmieta. Part **b**, bottom right, image courtesy of Corium Innovations.

the number of patents and clinical trials is continuously growing. We retrieved 138 clinical studies conducted between 2007 and 2022 from the [National Institute of Health ClinicalTrials.org](https://www.clinicaltrials.gov/) database by searching the keywords ‘microneedle’; out of the 138 studies, approximately one-quarter are phase II and later-stage studies (Table 2). The clinical indications include cosmetics, vaccinations, pain relief, local anaesthesia, ocular injection, cancer, osteoporosis, diabetes and diagnosis (Fig. 4a).

Hollow and solid MNs have been investigated for intradermal vaccination. In 2008, NanoPass Technologies used the MicronJet device (Fig. 4b, top) for intradermal delivery of influenza vaccines to 180 healthy adults (NCT00558649). The MicronJet device comprises an array of four silicon hollow MNs protruding from the plastic adaptor that can attach to syringes. Healthy participants were randomly allocated into three groups to accept an intramuscular full dose, an intradermal medium dose or an intradermal low dose of flu vaccine. The reduced dose intradermal vaccination resulted in a frequent but mild reaction on injection sites and a similar immunogenic response to intramuscular administration. The first-in-human study using dissolving MNs to administer inactivated influenza vaccine was conducted in 2017 (NCT02438423) with 100 participants enrolled and randomly allocated to four groups: intramuscular injection of inactivated influenza vaccine, inactivated influenza vaccine-loaded dissolving MN patch administered by a trained health-care worker, self-administered inactivated influenza vaccine-loaded dissolving MN patch, and placebo MN groups. This phase I study proved that MN vaccination was tolerated and resulted in comparable immunological responses to intramuscular

injection with the same dose. However, MN vaccination showed local skin reactions, which could be attributed to local immune responses occurring on the surface of the skin. MN-based vaccinations can potentially solve cold-chain storage issues, thereby facilitating vaccination in remote regions and during pandemics. Pharmaceutical companies have expressed interest in developing MN vaccination technologies; for example, in 2020, Merck announced a partnership with Vaxxas to develop an undisclosed vaccine MN product¹⁹⁵.

The approval and commercialization of MN patches for therapeutic purposes remain challenging. For example, Radius Health ceased its programme to develop an abaloparatide transdermal system (Abalo-TDS) for treatment of postmenopausal osteoporosis after a phase III trial. The Abalo-TDS achieved a clinical improvement in bone mineral density but was not comparable with subcutaneous injection. However, clinical studies continue on MNs for different therapeutic applications such as ocular drug delivery. For example, Clearside Biomedical developed the SCS Microinjector (Fig. 4b, middle); even though not unconditionally approved by the FDA, it was approved for suprachoroidal delivery through XIPIRE (triamcinolone acetonide injectable suspension).

The need for repeat insulin injections in diabetes management has motivated the development of painless transdermal delivery platforms for insulin, with clinical trials having started in 2008. Becton-Dickinson (NCT01061216 and NCT01557907) and NanoPass (NCT00602914) registered clinical trials to apply hollow injector systems to study the pharmacokinetics and pharmacodynamics of intradermally injected insulin. In 2013, Emory University conducted a small-size clinical trial of

12 participants using 900- μ m hollow MNs to transdermally deliver insulin, reporting rapid onset and lower pain compared to the subcutaneous insulin pump catheter (NCT00837512).

Several MN-based systems obtained FDA clearance and Conformité Européenne (CE) mark approval for painless blood sampling devices, including the TAP device (YourBio Health), OneDraw kit (Drawbridge Health) and the Tasso+ device (Tasso). By contrast, MN-based devices for the sampling and analysis of skin ISF are still under study. For example, MNs to collect and detect β -lactam antibiotics in skin ISF are in a phase I clinical trial (NCT03847610). Other clinical trials based on MNs have been registered, including the use of MNs to measure glucose (NCT02682056), medications for opioid use disorder (buprenorphine and methadone) and their metabolites (NCT05546229), and muscle nicotinamide riboside levels (NCT02712593).

Outlook

Several aspects should be considered in developing and commercializing MNs for therapeutic purposes (Box 1), including scale-up, cost-effective fabrication and product sterilization. Additionally, the device should work consistently regardless of location, race, sex, age or environment.

Commercializing a MN prototype depends on the cost-effective scale-up production of accepted quality based on pharmacopoeial standards. The large-scale fabrication of drug-free MNs is achievable as reflected by the cosmetic MN products on the market¹⁹⁶. Therapeutic

MNs can be favourable if the efficacy and cost of the treatment outperform other standard-of-care treatments. Sterilization is another fundamental requirement of medical products, using γ -rays, ethylene oxide or autoclaving based on the MN material and cargo loaded. Owing to the inherent sensitivity and administration requirement through the parenteral route, the biologics loaded in MNs are typically manufactured under aseptic conditions. Although aseptic processing is costly, it is essential to ensure product sterility.

Drug formulations should provide consistent dosage. The bioavailability of MN-based formulations depends on successful skin penetration and release of drugs or detachment of MNs from the patch baseplate into the skin. The bioavailability of active pharmaceutical ingredients may vary owing to the penetration efficacy, resulting in an uncontrollable therapeutic outcome¹⁹⁵. It is crucial to ensure the successful application of MN arrays in clinical trials by thumb pressing or applicators. Additionally, there are inter-individual and intra-individual variations (between different sites of the body) in skin thickness and mechanical properties. The penetration depth and drug release may differ even with the same pressing force and speed in MN application. Therefore, a deploying system will be needed to monitor and control the MN penetration depth. Skin penetration of MNs also affects their applications in transdermal biosensing. The inconsistent insertion depth may result in different compositions of extracted biofluids, and ISF can be contaminated by capillary blood or sweat. In addition, correlation studies between skin ISF and blood need to be

Box 1

Considerations for the clinical translation of MN technologies

Safety

Microneedle (MN) safety includes short-term and long-term safety. Short-term safety refers to the instant local and systemic responses of the body to the MN application such as bleeding, allergy, infection and rashes. Long-term safety is related to inflammation after MN insertion, the local and systemic responses of the body to periodic MN application, and local and systemic effects of the degraded products from MNs.

Stability

Drug-like biologics are often fragile and sensitive to external stress, such as high temperature, humidity and organic solvents, which might result in the denaturation, degradation and loss of drug potency. Excipients or cold-chain conditions can be used to protect from moisture, decrease degradation and stabilize the structures of biologics to improve drug stability.

Cost-effective mass production

Large-scale fabrication of MNs should pertain to pharmacopoeial standards and accepted quality control tests. It should also be cost-effective and consistent from batch to batch.

Sterilization

MN devices or drug–MN combinations have to be sterilized after manufacturing. Chemical and physical sterilization methods include

ethylene oxide, γ -rays and autoclaving. The chosen sterilization method should not change the function, integrity and appearance of the final product.

Packaging and storage

Some MN products have hygroscopic or oxidation problems. Thus, packaging with desiccants and nitrogen gas as well as immediate usage in humid conditions should be considered.

Deployment

Thumb pressing is a common method for MN application; however, other types of applicators have also been developed. There are intra-individual and inter-individual variations in skin thickness and mechanical properties. Even with the same pressing force and speed, the penetration depth and drug release could differ between individuals or different body parts. According to the specific indication, developing a smart deploying system that can monitor and control the MN penetration depth could be essential.

Regulations

The only official guidance for MN devices is from the FDA. In 2017 (draft version) and 2020 (final version), the FDA released their guidance for MN products²¹⁴.

more comprehensive to support the quantitative analysis of biomarkers, following the path of MN-based platforms for glucose measuring, especially with verified human-use devices²¹.

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Author contributions

All authors analysed the data, created figures and contributed to drafting, revising and reviewing the manuscript.

Competing interests

Z.G. is co-founder of Zenomics, ZCapsule and μ Zen Pharma. C.X. is co-founder of Greater Bay Biotechnology. The other authors declare no competing interests.

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